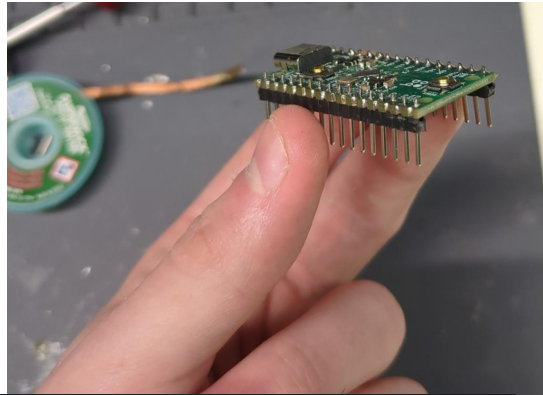


RISC V OS Spring Update 3

James Ryan, Evan Rosenfeld, Ant Nosaryev

Status

- Issues with attaching chip to board, applying the heat gun was not enough, had to also use a board heater. Managed to potentially compromise one chip, but one demo board is built
 - Work was spread out over last Friday and yesterday
- Serial output TBD – We know that UART is by default available from PB14/PB13 pins @ 1Mbps
 - Will be done today!
- Ran out of time to test



HotFudgeMuffin 2:21 PM

im putting this in the off topic chat since its not directly related to board development, imo

for those that put one of the boards together themselves, was it enough to put flux on the chip's pads, or did you also put paste on the pads? i've tried to heat the board and attach with just flux and the chip, but i had no luck. sorry if this is a really noobish question, i don't have much experience with board rework/assembly



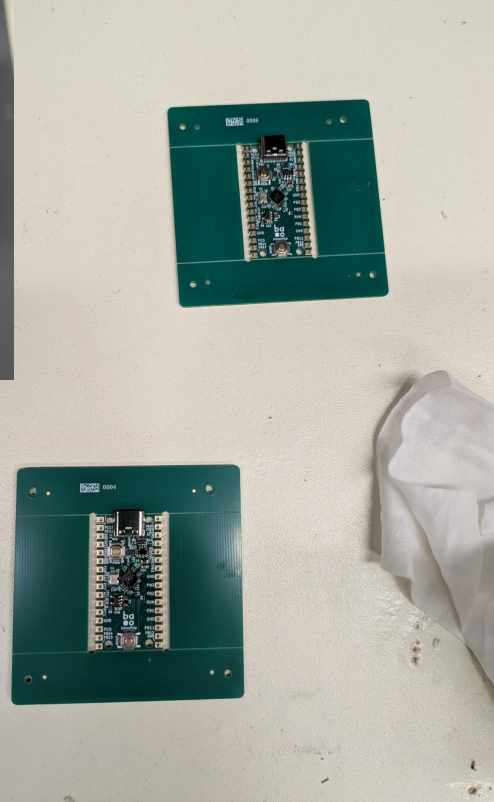
bunnie 3:17 PM

I just use flux only, no paste. you want the footprint as flat as possible, if that's the case, then the solder balls can do their trick. one problem is that the lead free stuff needs you to get the board to about 220C which is quite hot
i have to use a hot plate on the bottom *and* a hot air gun on top once the solder passes into its liquid phase you'll see the chip pull itself into place, i usually let the hot air run for a few more seconds after that. if you have very steady hands you can gently tap the side of the chip just to make sure all the solder balls took, but that takes a bit of practice not to tap it too hard and mess up everything



HotFudgeMuffin 3:23 PM

Ah, I was air gunning only. I'll try to set up a heat pad too



To do

- Set up common workstation or dev environment for easy use
 - There is a reproducible nix environment someone else has made which we can try to deploy on our own laptops
 - <https://github.com/betrusted-io/xous-core/pull/795>
 - Potential for open source contribution?
- Run a demo on board
 - <https://github.com/betrusted-io/xous-core/tree/dev/apps-dabao>
- Deliverable for next presentation: walk-through on the full system startup of bao1x with examples
 - <https://betrusted.io/xous-book/ch05-00-system-startup.html>

Next meeting is *after* spring break.